

4. The VMS shall allow for 2-way audio communication using amplifier/call station connected the IP cameras in the field without any need of audio cabling from camera to control room

5.5.1.12.7 Video Analytics

5.5.1.12.7.1 General Requirements:

1. The Video Analytics shall be designed to provide Intelligent Video Analysis for 24/7 surveillance with support for devices from different vendors.
2. Support any architecture namely distributed, centralized and hybrid
3. Support system openness without using any proprietary format
4. Support commercial-off-the-shelf computing hardware without the need of any proprietary hardware
5. Able to produce reliable analytics at lower resolutions like 4CIF resolution in order to save the computation
6. Able to process at variable resolution and frame rate when if necessary
7. It shall support open platform Video Management System (VMS).
8. It shall provide ONVIF device discovery
9. It shall get video from camera or VMS and send alarms to VMS to be viewed in VMS client
10. It shall stream the Analytics Video to VMS using open interface protocol like ONVIF.
11. It shall support multiple regions of analytics on single video feed.
12. It shall support multiple features to be enabled for each of the regions.
13. It shall support feature based scheduling so that that alarms can be enabled or disabled for a certain period of time.
14. It shall support both Virtual line and Virtual area based features. The virtual area can be of any shape and can be bound by at least 10 end points.
15. It shall support both indoor and outdoor environment.
16. It shall support setting of minimum and maximum object size for detection.
17. It shall support masking of area in a view.
18. It shall support object masking.
19. It shall support analytics capability to run both on server as well as edge (on camera).
20. It shall support simultaneous running of different features both on edge as well as server for same camera
21. It shall support camera independent licensing
22. The VMS shall provide a centralized camera tampering detection solution in real-time by automatically identifying tampering to ensure video image capture and integrity. The solution sends an alert when the following potential tampering is detected:
 - Scene too bright — e.g. flash light, direct sun, laser pointer that is pointed at the camera, causing it to become over saturated.

- Scene too dark — not enough light to see a clear image, if camera is covered.
- Camera is covered or blocked — if something is blocking or partially blocking most of the camera's field of view.
- Camera redirection detection — if camera is redirected from its' initial position of field of view (FOV).
- Unfocused or blurred view — if the camera was sprayed with rain or its focus changed.

23. Automatically search through the recorded video for the original appearance of an object in a scene.

5.5.1.12.7.2 Suspicious incident detection

1. It shall detect person loitering in a virtual area for more than a pre-defined period.
2. It shall detect crowd assembling in a pre-defined area. The count for the crowd determination should be pre-defined. It shall be able to provide live crowd count.
3. The VA shall support dense and sparse crowds for crowd counting and crowd flow detection
4. The VA shall detect object left out or abandoned in a virtual area by a person beyond a certain pre-defined period.
5. The VA shall detect object removed by a person beyond a certain pre-defined period.
6. The VA shall detect counter flow of people (such as people moving in a wrong way)
7. The VA shall be able to efficiently locate and track a specific person across time and location to minimize search time from hours to minutes when time is of the essence.
8. The VA shall be able to track individuals' movements from location to location and access all relevant associated VMS recordings
9. The VA shall support track individual algorithm to display tracked individuals path and directions on the map.
10. The VA shall support creating a composite (a human-like figure) of the suspect based on eye-witness' description. Many different options should be made available for describing hair color and style, shirts, trousers, patterns, etc.

5.5.1.12.7.3 Other features

1. It shall be able to stabilize the video when camera is shaking (such as, due to wind) and shall be able to stream the stabilized video to VMS.
2. Ability such that alerts can be searched and categorized based on this information.
 - i. Timestamp (date & time)

- ii. Alert Name
 - iii. Alert Type
 - iv. Alert Location
 - v. Text Description
 - vi. Associated Region
1. It shall provide video summary of all the alarms.
 2. It shall provide reporting option to export reports of alarms in PDF, EXCEL and Image formats and also option to schedule it.
 3. It shall support email and FTP of alarm data and also option to schedule it.
 4. It shall be able to provide comparison reports for different months and year

5.5.1.12.8 Standardized Signs for CCTV Camera Locations

It is necessary that the CCTV Camera locations put some standardized signs informing the public of the existence of CCTV cameras. This will bring about the transparency on installation of CCTV cameras and no one would be able to later complaint for breach of privacy. Following tables give draft specifications for the signage to be put at the camera locations.

#	Item	Specifications
1	Size	Board Width = 8" / 12" (For type A and B) Board Width = 12" / 18" / 24" (For type C and D)
2	Plate Material	Corrosion resistant Aluminum Alloy as per IRC 67:2001 (Code of Practice for Road signs)
3	Plate Thickness	Minimum 1.5 mm
4	Retro-Reflective sheeting for sign-plate	Weather-resistant, having colour fastness
5	Other Specifications	As per IRC 67:2001 (Code of Practice for Road signs)
6	Mounting	Can be mounted on wall or pole (appropriate mounting brackets to be provided)
7	Design	As per following signage diagrams

5.5.1.12.9 Fixed Box cameras

#	Parameter	Minimum Specifications
		- Support 802.1x (Port based network access control). - Support for MAC filtering.
6.	PoE Power per port	Sufficient to operate the CCTV cameras connected
7.	Rating	IP 30 or equivalent Industrial Grade Rating (This is not essential if the switch is placed in equivalent or better junction box / enclosure)
8.	Operating Temperature	0 – 50 degrees C or better

5.5.1.12.12 Face Recognition System

Face Recognition System (FRS) shall be designed for identifying or verifying a person from various kinds of photo inputs from digital image file to video source. The system shall offer logical algorithms and user-friendly, simple graphical user interface making it easy to perform the facial matching.

The system shall be able to broadly match a suspect/criminal photograph with database created using photograph images available with Passport, CCTNS, and Prisons, State or National Automated Fingerprint Identification System or any other image database available with police/other entity.

The system shall be able to:

- i. Capture face images from CCTV feed and generate alerts if a blacklist match is found.
- ii. Search photographs from the database matching suspect features.
- iii. Match suspected criminal face from pre-recorded video feeds obtained from CCTVs deployed in various critical identified locations, or with the video feeds received from private or other public organization's video feeds.
- iv. Add photographs obtained from newspapers, raids, sent by people, sketches etc. to the criminal's repository tagged for sex, age, scars, tattoos, etc. for future searches.
- v. Investigate to check the identity of individuals upon receiving such requests from.

vi. Enable Handheld mobile with app to capture a face on the field and get the matching result from the backend server.

The facial recognition system shall be enabled at cameras identified by the Authority. These cameras identified shall be installed at critical locations finalized by authority

The facial recognition system in offline mode shall be provided by the SI in line with the requirement specified in the RFP.

The detailed functional requirement specification of the facial recognition system is provided in subsequent sections of this RFP.

5.5.1.12.12.1 Face Image Data Standard

Manual Facial recognition is not sufficient currently for de-duplication. . Computer based face recognition has reasonable accuracy under controlled conditions only. Hence for de-duplication purposes, other biometrics like finger print/iris image is also recommended.

With the objective of interoperability among various e-Governance applications, the face image data standard for Indian e-Governance Applications will adopt **ISO /IEC 19794-5:2005(E)**. While the ISO standard is broad to cover all possible applications of computer based face recognition and human visual inspection, this standard is more restrictive, as it is limited to human visual inspection.

The ISO standard specifications are tailored to meet specific needs of civilian e-Governance applications by specifying certain prescriptive values and best practices suitable in Indian context.

Standard	Description
ISO /IEC 19794-5:2005(E)	This standard includes capture and storage specifications of face images for human visual inspection and verification of the individuals in Indian E-Governance applications. It specifies a format to store face image data within a biometric data record compliant to the Common Biometric Exchange Formats Framework (CBEFF), given in ISO 19785-1. It also includes best practices recommended for implementation of the specifications in different categories of e-Governance applications. The scope of this standard includes: - o Characteristics of Face Image capturing device - o Specifications of Digital Face Image & Face Photograph Specifications intended only for human visual inspection and verification

5.5.2 Smart solutions with Artificial Intelligence at Edge Devices:

Along with the components of the CCTV Surveillance system, the SI would be responsible to integrate the following services with the CCTV Surveillance system to build an infrastructure for a Smart city system in the Chennai Area having the functional requirement to be cater through Artificial Intelligence at edge devices with continuous and deep learning capabilities for real time response and monitoring.

These use cases are to be implemented using Artificial Intelligence through various cameras, sensors etc at the edge/field devices with continuous learning capabilities, Following use cases are to be part of implementation but not limited to:

- Graffiti and Vandalism detection
- Debris and Garbage detection
- Attendance of sanitation workers on site by face recognition
- Sweeping and cleaning of streets/bins before and after
- Garbage bin, cleaned or not
- Litter detection
- Tracking of garbage truck movement and Quantity of garbage dumped at dumpsite
- Detection and Recognize the pattern of demonstration and conflicts in crowd
- Detection and classification of human, animal and vehicle
- Safety: Detection and classification based on :
 - o Behavioural Biometry : Identification through multiple behaviour
 - o Parking violation
 - o Speeding vehicle
 - o Accident detection
 - o Loitering detection
 - o Person climbing barricade
 - o Person collapsing
 - o Person/Face recognition
 - o Gesture recognition : Identification through gesture change
- 'Vehicle of interest' tracking by colour, speed, number plate
- Helmet detection on two wheeler
- Unwanted/ banned vehicle detection
- Wrong way or illegal turn detection
- Toilet cleaning by detection of smell etc
- Water quality sensors at District Metered Area level
- Environmental condition detection
- Air quality detection

Edge based Analytics min specification for functional delivery:

- Deliver 5.5 teraFLOPS of FP16 performance or better
- Support Wi-Fi, Bluetooth, 3G/LTE connectivity options
- Have 100Mb/1GbE management network link
- Support 1GbE (via RJ45) or 10GbE (SFP+)

- Have operating temperature range of 0 degrees to 55 degrees to withstand outdoor operating environment and as per city conditions

Artificial Intelligence with Continuous Learning & Improvement system Specification:

- Deliver processing units' performance of 1 petaFLOPS on FP16 or better
- Have software tools for achieving the following tasks- Resource allocation, queueing of jobs, performance monitoring and creating software containers
- Support commonly used Deep Learning based AI frameworks like TensorFlow, CNTK etc.
- Have minimum 512GB system memory per system or better
- Have dual 10GbE and 4 IB EDR per system or better
- Min power consumption per system.
- Have dual 20-core Intel Xeon E5-2698 or better per system.
- At least Support parallel computing architecture.
- Support software libraries for continuous learning and improvement for betterment of Intelligent video analytics software installed in edge/field devices using Deep Learning based AI methodologies.